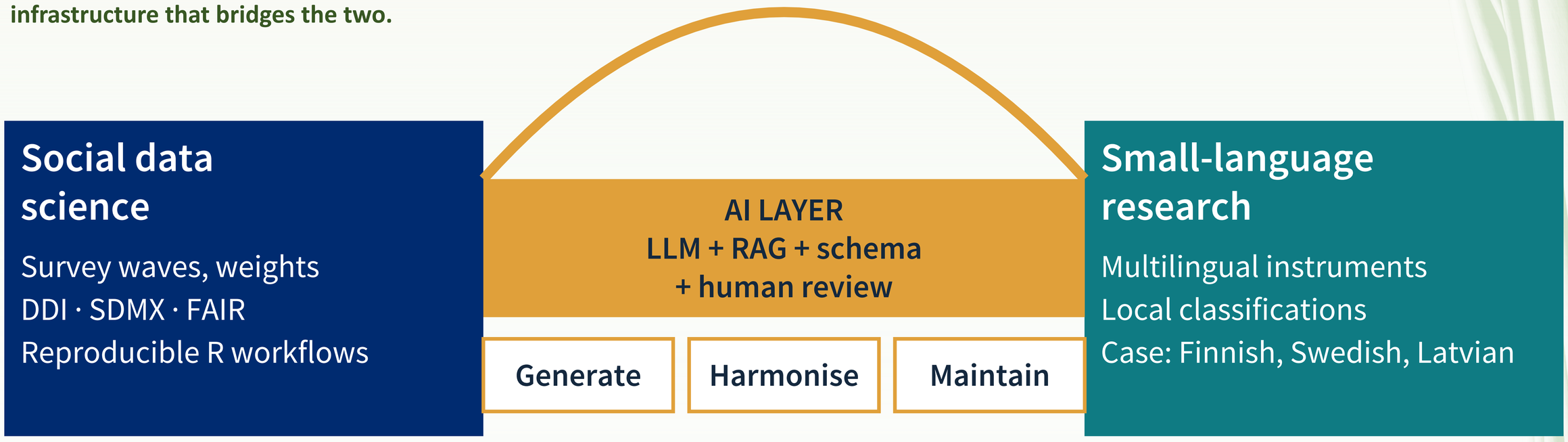
UNIVERSITY OF HELSINKI
FACULTY OF SOCIAL SCIENCES

The Data Bridge

Connecting Social Data Science and Small-Language Research with AI

SurveyAI · DARIAH-FI · Centre for Social Data Science

Survey data is one of the richest structured sources of Finnish- (and Swedish-)language social data. It is only reusable when it is well documented, and the AI tools used to document and analyse it are built for English. **We build open infrastructure that bridges the two.**



1 The bottleneck

- Documentation is a manual, error-prone add-on
- DDI and SDMX exist, but adoption is uneven
- Poor metadata means lost reuse and lost knowledge
- Lack of FAIR means increased costs and survey burden to small populations if new surveys needed
- Lack of reuse means underrepresentation in international comparisons

2 Multilingual challenge

- Finnish, Swedish and Latvian versions must stay consistent in content
- Local classifications deviate from standard vocabularies
- Legacy documentation is not AI-ready
- Semantic interoperability is difficult to achieve

3 Pipeline: R verifies, the LLM writes

Data input CSV or SPSS → **R summary** labels, counts → **LLM drafts** descriptions → **roxygen2** documented data → **pkgdown** docs site → **Codebook** with reliabilities

The LLM never guesses from raw data: it receives a verifiable summary produced by R.

4 Guardrails

- **RAG:** Prior waves ground the output
- **Schema:** Output checked against DDI
- **Humans:** Experts approve every release
- **Risks:** Hallucination, silent errors

5 Take-home

- A demanding but productive test case
- Guardrails for AI keep the risks manageable
- Towards FAIR, shared infrastructure
- **Next:** FI / SV / LV tests

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